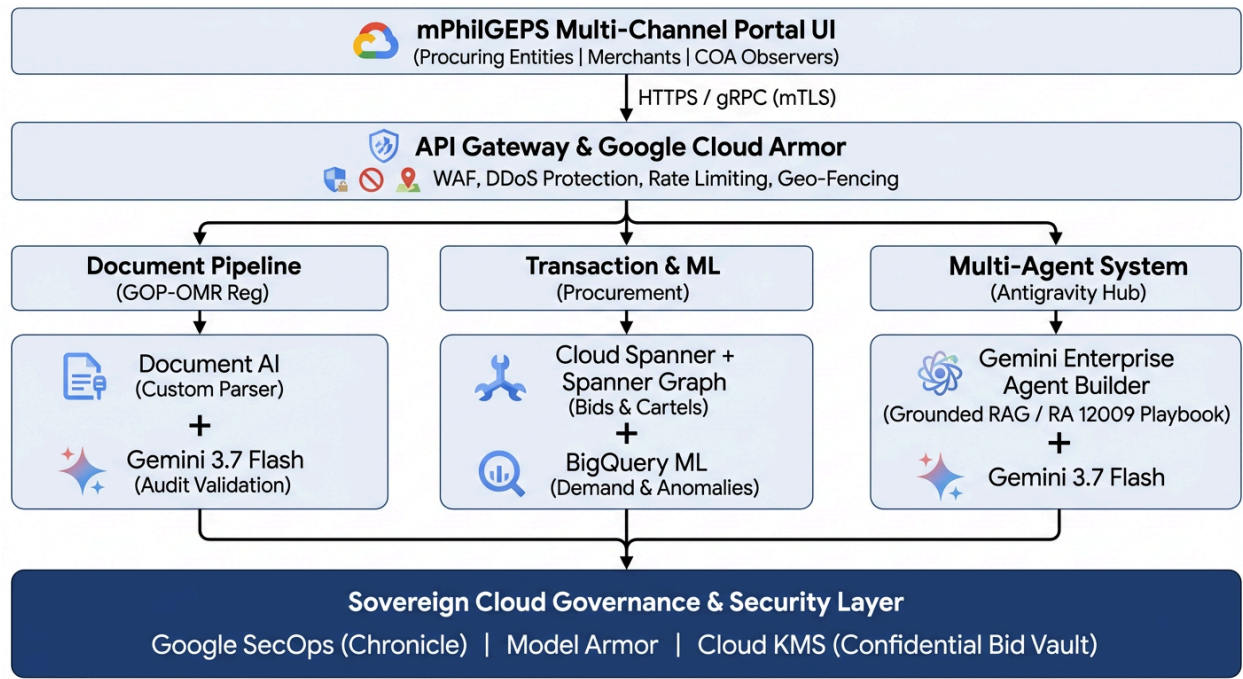


Google Cloud’s modern AI stack provides mPhilGEPS Phase 2 with a unified, sovereign platform that directly operationalizes the mandates of RA 12009 (New Government Procurement Act)—spanning automated UNSPSC taxonomy mapping, document verification, multi-agent conversational workflows, and graph-based anti-collusion detection.

 Key Takeaway & Architecture Principle

The Core Architectural Thesis: Rather than bolting disparate third-party AI APIs onto legacy relational databases, mPhilGEPS Phase 2 should leverage **in-database AI processing (BigQuery ML and Spanner Graph)** paired with **Gemini 3.1 Pro (heavy reasoning)** and **Gemini 3.7 Flash (sub-second transactional inference)**. This architecture maintains end-to-end data residency under RA 10173, isolates unopened bids cryptographically, and scales seamlessly to support peak loads exceeding **5,500 concurrent users**.



1. AI-Powered Procurement Insights & Hierarchical UNSPSC Classification

Procuring entities frequently submit requisitions and Annual Procurement Plans (APPs) with free-text, colloquial, or non-standard item descriptions. Manually mapping tens of thousands of line items into the 8-digit **United Nations Standard Products and Services Code (UNSPSC)** framework creates administrative latency and corrupts procurement reporting.

Technical Implementation Details

- **Dual-Pass Extraction & Classification:**

- **Pass 1 (Semantic Normalization): Gemini 3.7 Flash** parses raw item titles, specifications, and unit measurements, stripping agency jargon and brand specifics into standardized semantic definitions.
- **Pass 2 (Hierarchical Enforcement via Gemini 3.1 Pro):** When complex engineering items or bundled tenders are processed, **Gemini 3.1 Pro** is invoked with strict **Structured Outputs (JSON Schema)**. It matches against the official UNSPSC hierarchy:

Segment (XX)→Family (XX)→Class (XX)→Commodity (XX)

- **Schema Definition & Deterministic Output:**

JSON

```
{
  "unspsc_code": "43211507",
  "hierarchy": {
    "segment": "Information Technology Broadcasting and Telecommunications",
    "family": "Computer Equipment and Accessories",
    "class": "Computers",
    "commodity": "Desktop computers"
  },
  "confidence_score": 0.985,
  "extracted_specifications": {
    "form_factor": "Small Form Factor (SFF)",
    "processor_class": "Enterprise 16-Core",
    "memory_gb": 32
  }
}
```

-
- **Vendor-Tender Matching (Vertex AI Vector Search):**
 - Historical awards and merchant merchandise profiles from the **Government of the Philippines Official Merchant Registry (GOP-OMR)** are embedded using `text-embedding-005` (768 or 1536 dimensions) and indexed in Vertex AI Vector Search.

- When a new Invitation to Bid (ITB) is posted, vector search computes nearest neighbors within milliseconds to proactively push notifications to certified suppliers, maximizing competitive bidding.

2. Predictive Analytics & Public Demand Forecasting

RA 12009 mandates strategic procurement planning to prevent artificial budget exhaustion and optimize bulk purchasing across departments (e.g., PS-DBM bulk buys).

Technical Implementation Details

- **In-Database Time-Series Forecasting (BigQuery ML):**
 - By training directly on multi-year transactional datasets, **BigQuery ML (ARIMA_PLUS)** avoids exporting sensitive government fiscal data outside the database boundary.
 - The model automatically accounts for seasonality, Philippine national holidays, fiscal year-end spending spikes (Q4 rushes), and inflation trends.
- **SQL Pipeline Example:**

SQL

```
CREATE OR REPLACE MODEL `mphilgeps_analytics.app_demand_forecast`  
OPTIONS(  
  model_type = 'ARIMA_PLUS',  
  time_series_timestamp_col = 'procurement_date',  
  time_series_data_col = 'total_expenditure',  
  time_series_id_col = 'unspsc_commodity_code',  
  holiday_region = 'PH'  
) AS  
SELECT  
  procurement_date,  
  unspsc_commodity_code,  
  total_expenditure  
FROM  
  `mphilgeps_dw.historical_purchase_orders`;
```

-
- **MLOps via Vertex AI Pipelines (Kubeflow-based):**

- Tracks feature drift when major macroeconomic disruptions or executive budget realignments occur.
- Automatically schedules model re-training and updates automated replenishment alerts for standard office supplies and medical stockpiles.

3. Automated Document Verification & Merchant Accreditation (GOP-OMR)

Merchants applying for **PhilGEPS Platinum Membership** must submit extensive legal, financial, and technical credentials. Manual verification by BAC secretariats creates backlogs spanning weeks.

Verification Architecture Matrix

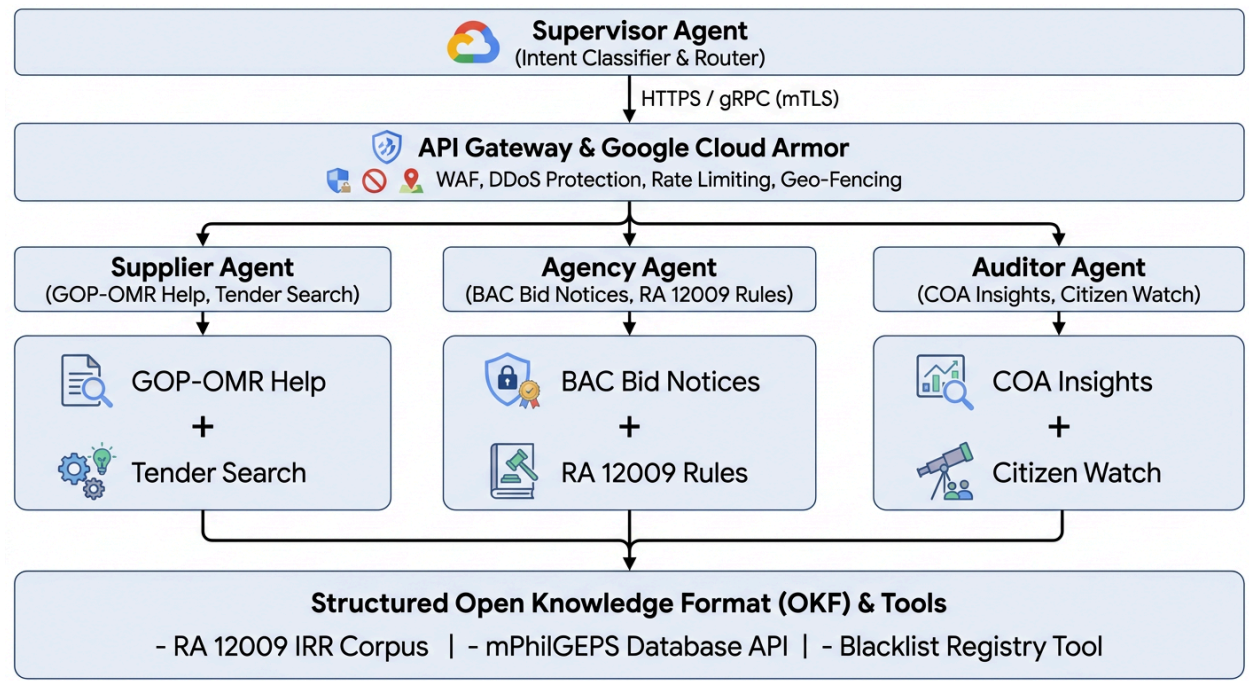
Document Type	Extractor Engine	Target Fields for Verification	Cross-Validation Rule Check
SEC General Information Sheet (GIS)	Document AI Custom Extractor (CDE)	Corporate Name, SEC Reg No., Board of Directors, Stockholders, Equity Breakdown	Validates against 60/40 Filipino ownership threshold; flags shared officers with other bidding firms.
DTI Business Registration	Document AI CDE	Business Name, Owner Name, Certificate Scope, Validity Period	Verifies business status and cross-checks owner against GPPB debarment lists.
BIR Tax Clearance	Document AI Form Parser	Tax Clearance No., Certificate Date, Expiration Date, TIN	Validates certificate authenticity via cryptographic barcode and ensures certificate is not expired.
PCAB Construction License	Document AI CDE	License Number, Classification, Category (AAA/AA/A), Expiry Date	Checks that contractor capacity matches or exceeds the Approved Budget for the Contract (ABC).

Audited Financial Statements (AFS)	Document AI Layout Parser + Gemini 3.1 Pro	Current Assets, Current Liabilities, Net Worth	Automatically calculates Net Financial Contracting Capacity (NFCC) : $NFCC = [(Current\ Assets - Current\ Liabilities) \times 15] - Value\ of\ Outstanding\ Works$.
---	--	--	--

- Audit Decisioning Engine: Gemini 3.7 Flash** parses the combined JSON output from Document AI, verifies mathematical accuracy of the NFCC calculation, checks expiration dates against the tender submission deadline, and marks the application **APPROVED**, **FLAGGED**, or **REJECTED_WITH_REASON**.

4. Multi-Agent Conversational Ecosystem (Google Antigravity & Agent Builder)

Rather than implementing a single generic chatbot, mPhilGEPS requires an enterprise **multi-agent orchestration framework** governed by **Google Antigravity**.



Specialized Agent Roles

- The Merchant Onboarding & Bidding Agent:**
 - Contextually guides bidders through submitting technical and financial bids.

- Provides real-time pre-flight bid checks (e.g., *"Your Omnibus Sworn Statement is missing Annex A"*).
 - 2. **The Bids and Awards Committee (BAC) Advisor Agent:**
 - Grounded in the **RA 12009 Implementing Rules and Regulations (IRR)** and GPPB Standard Bidding Documents.
 - Assists government procurement officers in formulating evaluation criteria, specifying technical requirements without anti-competitive brand names, and calculating procurement timelines.
 - 3. **The Public Transparency & COA Auditor Agent:**
 - Built for Commission on Audit (COA) personnel and civil society monitors.
 - Translates natural language into governed queries (e.g., *"Compare bid price variations for flood control projects across Region III in 2025"*).
 - **Grounding via Structured Open Knowledge Format (OKF):** Standard RAG often yields inaccurate answers on statutory references. Knowledge chunks are indexed with strict metadata tags (section, subsection, amendment date, applicability), ensuring citation accuracy with zero hallucination.
-

5. Graph-Based Anti-Collusion, Bid-Rigging & Anomaly Detection

Cartels and anti-competitive bidding rings often coordinate their submissions using phantom competitors, cover bidding, and circular rotation. Standard relational tables struggle to identify deep multi-hop links across entities.

Technical Implementation Details

- **Spanner Graph Model:**
 - **Nodes:** Bidder, Director, Authorized_Signatory, Bank_Guarantor, IP_Address, Physical_Address, Tender.
 - **Edges:** SUBMITTED_FOR, SERVES_ON_BOARD, REGISTERED_WITH_ADDRESS, UTILIZED_NETWORK, FINANCIALLY_BACKED_BY.
- **Graph Pattern Detection (GQL in Spanner):**

SQL

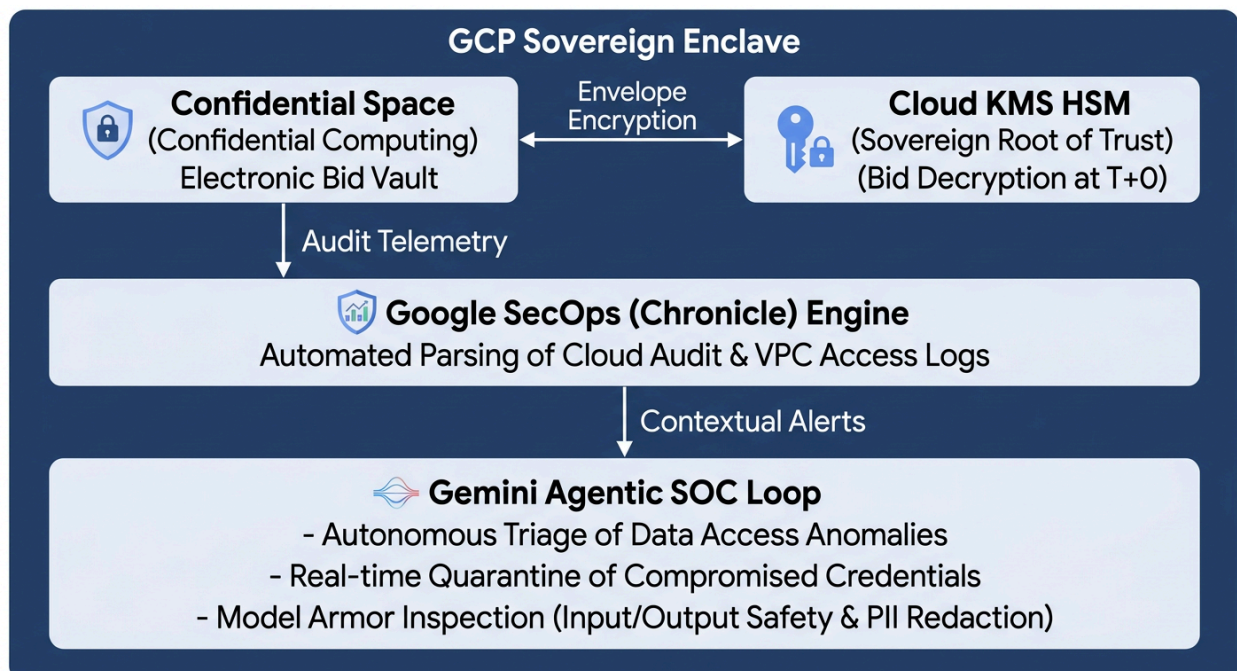
```
GRAPH MPhilGepsProcurementGraph
MATCH
(b1:Bidder)-[:SERVES_ON_BOARD]->(d:Director)<-[:SERVES_ON_BOARD]-(b
2:Bidder),
      (b1)-[:SUBMITTED_FOR]->(t:Tender)<-[:SUBMITTED_FOR]-(b2)
WHERE b1.id != b2.id
```

```
RETURN b1.company_name, b2.company_name, d.full_name,  
t.reference_number;
```

-
- **Anomaly Detection Metrics:**
 - **Bid Rotation Index:** BigQuery ML clustering algorithms analyze historical tender wins within specific geographic zones. When firms alternate winning bids with statistical precision, the algorithm assigns a high **Collusion Probability Index (CPI)**.
 - **Benford's Law & Line-Item Pricing Variance:** Detects bids where unit cost distributions deviate from natural procurement distributions, or where competing bidders have identical line-item decimal anomalies.

6. Sovereign Cloud Security, Agentic SOC & Zero-Trust Governance

Procurement data before bid opening is classified under Philippine law as confidential and market-sensitive. Leaks can compromise multi-billion-peso national infrastructure programs.



Security & Compliance Controls

- **Confidential Space Bid Vault:** Electronic bids are encrypted client-side and stored in an isolated enclave running on **Confidential VMs**. The private decryption key is managed in Cloud KMS and locked until the BAC chairperson and COA observers simultaneously sign the digital release token at the bid opening timestamp.
 - **Autonomous Agentic SOC (SecOps + Gemini):**
 - Ingests tens of thousands of EPS events per second into **Google SecOps (Chronicle)**.
 - An automated security agent detects brute-force enumeration, abnormal database read queries on tender pricing tables, and access attempts originating outside whitelisted sovereign boundaries.
 - **Model Armor & Data Privacy (RA 10173):**
 - Inspects every prompt and inference output.
 - Masks Tax Identification Numbers (TIN), personal mobile numbers, and bank account details from public-facing auditor assistants.
 - Mitigates prompt injection attacks aimed at altering agent bidding advice or leaking internal BAC deliberations.
-

Non-Functional Requirements & Sizing Parameters

- **Concurrency Thresholds:** Scaled to support baseline traffic of **3,500 concurrent sessions** and peak bid-closing surges of **5,500 concurrent sessions** via Google Kubernetes Engine (GKE Autopilot) with Cloud Spanner autoscaling.
- **Latency Budget Allocation:**
 - Interactive UI Searches & Portal Navigation: **< 1,000 ms** (Powered by Cloud CDN and Gemini 3.7 Flash).
 - Batch Document Parsing (SEC/GIS/BIR): Asynchronous Pub/Sub pipeline with notification callbacks (**< 15 seconds per 20-page document**).
 - High-Complexity Collusion Graph Queries: Scheduled nightly ETL or on-demand BAC pre-award evaluation jobs (**< 60 seconds**).